



Research Article

The Significance of Laboratory Findings in Post-Traumatic Dementia

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Summary

Introduction: Head trauma cases can be converted to dementia during the follow-up period. This condition is called “post-traumatic dementia (PTD)”. Dementia can be classified according to clinical stages. The aim of this study is to determine whether there is a difference between clinical stages of PTD and laboratory, neuroimaging, neuropsychological tests.

Method: The patients with PTD were categorized into three groups: mild, mild-moderate and moderate stages. Mini mental state examination (MMSE), verbal fluency, frontal assessment battery (FAB), clock drawing test (CT), vitamin B12, thyroid-stimulating hormone (TSH), electroencephalography (EEG) and brain magnetic resonance imaging (MRI) results of the groups were documented. Findings were compared among the groups. SPSS 20.0 package programme was used to make comparisons.

Results: There were classified into mild (n:51), mild-moderate (n:21) and moderate stages (n:19) PTD. There was no significant difference between stages on verbal fluency, FAB, CT, vitamin B12, TSH, EEG, MRI results ($p>0.05$) except for MMSE ($p=0.01$). Despite no significantly correlation was found between stages and MRI, the rate of mild stage dementia patients with temporal lesions was less than the other dementia groups.

Conclusion: In the present study; significantly difference was found between stages of dementia and MMSE. We also found out that the rate of mild stage dementia patients with temporal lesions was less than the other dementia groups. These findings suggest that possible use of combined MRI and MMSE could play a role in staging of PTD.

Key words: Trauma, Dementia, Laboratory

Posttravmatik Demansda Laboratuvar Bulgularının Önemi

Özet

Giriş: Kafa travma öyküsü olan olgular klinik takipleri boyunca demans kriterlerini karşılayabilirler. Bu duruma posttravmatik demans (PTD) denir. Demans ise klinik şiddetine göre evrelere ayrılır. Bu çalışmada PTD olgularının evreleriyle laboratuvar, görüntüleme ve nöropsikolojik testler arasındaki farklılığın saptanması amaçlanmıştır.

Method: Hafif, hafif-orta ve orta olmak üzere üç gruba ayrılan PTD olgularının minimal durum testi (MMSE), verbal akıcılık, frontal değerlendirme bataryası (FAB), saat çizme testi (ST), vitamin B12, tiroid stimulan hormon (TSH), EEG ve beyin manyetik rezonans görüntüleme (MRG) sonuçları dökümanite edilmiştir. Veriler SPSS 20.0 paket programı kullanılarak gruplar arasında karşılaştırılmıştır.

Bulgular: Hafif evre (n:51), hafif-orta (n:21) ve orta evre (n:19) PTD tanısı almış 3 grubun verileri karşılaştırılmıştır. Verbal akıcılık, FAB, ST, vitamin B12, TSH, EEG, MRG açısından gruplar arasında farklılık gözlenmezken ($p>0.05$), MMSE açısından gruplar arasında farklılık gözlenmiştir ($p:0.01$). MRG'de anlamlı farklılık gözlenmemesine rağmen, temporal lezyonu olan hafif evre demans olgularının oranı diğer gruplara göre düşük saptanmıştır.

Sonuç: Bu çalışmada demans evreleri ile MMSE arasında anlamlı farklılık gözlenmiştir. Ayrıca temporal lezyonu olan hafif evre demans olgularının oranı diğer gruplara göre düşük saptanmıştır. Bu bulgular göz önüne alındığında MMSE ve MRG'nin birlikte kullanımının PTD evrelemesinde rol alabileceği düşünülmüştür.

Anahtar Kelimeler: Travma, Demans, Laboratuvar

INTRODUCTION

Head trauma is a common health problem in developing countries, the most common causes of which are traffic accidents, followed by falls from heights and sports injuries⁽¹¹⁾. Head trauma can lead to multiple neurological, psychiatric and cognitive deficits^(3,6,16,19,21). Due to abrupt acceleration and deceleration during a traumatic injury, "diffuse axonal damage" may occur in the orbitofrontal-temporal cortex, corpus callosum, superior cerebellar peduncle and the basal ganglia⁽¹⁷⁾. The frontal and temporal lobes are the most sensitive areas. Emotional lability, behavioral changes, and impairments in attention, planning, orientation, executive function and memory can be observed due to damage to the brain. Sequel cognitive loss may be observed approximately 2-5 years after the trauma, which can deteriorate the daily life⁽¹⁾.

Dementia is a clinical condition arising due to cognitive loss, which affects the daily activities. In the period following the trauma, 5 % of the cases are diagnosed as dementia⁽³⁾. This condition is called post-traumatic dementia (PTD). PTD could be confused with major depression and post-traumatic stress disorder^(2,9). Therefore, objective tests are needed to evaluate the cases. Imaging of PTD patients can be abnormal or within normal ranges^(3,4,7,15). The inconsistency between the clinical picture and the imaging is due to diffuse microscopic lesions. Therefore, studies have been conducted on structural and functional imaging. In some studies, a small correlation between neurocognitive tests and imaging has been determined; however, no correlation has been

determined in some other studies^(4,7,15). No research has been carried out on the correlation between the imaging and the clinical stages of PTD in the literature. The aim of this study was to determine whether there was a difference between post-traumatic dementia stages and the laboratory and neuropsychological tests.

MATERIAL AND METHODS

Subjects

Patients who had PTD were included in the study between 2011 and 2013. The study was carried out at the Neurology Clinic of Cukurova University Hospital. Patients who were illiterate, who were in the first year of trauma, who had other degenerative forms of neurological (Alzheimer dementia, Parkinson's disease etc.) or systemic (hypothyroid disease, connective disease, etc.) disease, alcohol abuse, current use of medications known result in cognitive decline (antipsychotic, antidepressants, etc.) were excluded from study. All patients were evaluated by three neurologists experienced in dementia patients. The DSM-4 criteria were used for the diagnosis of dementia. The clinical dementia rating scale¹⁸ and the global deterioration scale⁽²⁰⁾(GDS) were used for the staging of dementia. The patients were categorized in three groups according to GDS scores, mild (GDS; 4), mild-moderate (GDS; 5) and moderate (GDS; 6) respectively.

The patients' gender, age, etiology of trauma, duration between trauma and diagnosis of dementia (duration), symptoms, neurological examination, neurocognitive tests, levels of vitamin B12 and thyroid-stimulating hormone (TSH), electroencephalography (EEG) and the brain magnetic resonance imaging (MRI)

were documented. The clinical and laboratory findings were compared according to the dementia stages.

Neurocognitive Tests

Mini mental state examination⁽¹¹⁾ (MMSE), verbal fluency⁽¹³⁾, frontal assessment battery⁽⁵⁾ (FAB) and clock drawing test⁽¹⁴⁾ were performed by a neurologist experienced in neurocognitive tests.

Laboratory

MRI

Imaging was performed with 1.5 Tesla scanner and sequences of axial and coronal dual T2 fast relaxation fast spin echo, axial T2 fluid-attenuated inversion recovery (FLAIR) and axial T1 MRI.

EEG

EEGs were performed with a 14-channel analogue device. Electrodes were placed in the 10-20 system position, within the frequency range of 0.03-70 Hz, at an amplitude sensitivity of 100 μ V. Hyperventilation and photic stimulation were applied as the activation method. The results were evaluated as normal or abnormal [epileptic activity (spike and wave, sharp and wave) or nonepileptic abnormality (diffuse ground arrhythmia)].

Serum Levels

Laboratory analyses including thyroid-stimulating hormone (TSH) and vitamin B12 were performed. Blood samples were collected between 7:00 and 8:00 AM and placed in a cooled container and immediately transported to the laboratory where the plasma was separated by centrifugation. The blood samples were studied by immunofixation methods in the "the Beckman Coulter DXI 800".

Statistical Analysis

The SPSS 20 package was used in the statistical data analysis. Categorical measurements were summarized in figures and percentages, while standard deviations (median and minimum/maximum where

required) were used for numerical measurements. The Chi Square test statistics were used in inter-group comparisons of categorical measurements and the Mann Whitney U test for inter-group comparison of numerical measurements that showed non-normal distribution. For the general comparison between more than two groups whose numerical measurements showed non-normal distribution, the Kruskal-Wallis test was used. For the conditions found to be significant in these comparisons, the Mann Whitney U Test with Bonferroni adjustment was used in the paired comparison of groups. To evaluate the correlations between basal measurements, Pearson Correlation Coefficient or Spearman Rank Correlation Coefficient was used depending on whether the statistical hypotheses were fulfilled or not. Statistical significance was set as 0.05 in all tests.

RESULTS

A total of 91 patients with PTD (13 % female) were included in the study. Fifty one (56%) of them met the criteria in the mild stage, the remainder in the mild-moderate (23%), and moderate (21%) stages. Of these patients, the head trauma had resulted from a traffic accident in 64, and remainder had suffered from other head trauma causes (falling down, etc.). The demographic and PTD related characteristics for the 91 subjects have been demonstrated in Table 1.

Cognitive examination revealed amnesia, loss of motivation and concentration in 68 patients, personality and behavioral changes in 23 patients. Neurological examination revealed hemiparesis in 6 of the patients.

The verbal fluency, frontal assessment battery and clock drawing test scores were not significantly different between the stages of PTD groups, while MMSE was significantly different between the groups ($p:0.01$) (Table 1). Furthermore, no difference was found between PTD stages

and the vitamin B12, TSH, MRI, EEG findings (p: 0.20; 0.97; 0.87; 0.31, respectively) (Table 2). Despite no significant difference having been found between the stages and MRI, the rate of mild stage dementia patients with temporal lesions was lower than the other dementia groups (Table 2).

On evaluation of the correlation analyses, there was a negative correlation between MMSE and GDS ($p<0.001$, $r: -0.474$), and no correlation was found between other neurocognitive, laboratory tests and GDS ($p>0.05$).

Table 1: Comparison of Data According to Stages of Dementia

	Mild (n:51, %56)	Mild-Moderate (n:21, %23)	Moderate (n:19, %21)	p
Female (n)	9	2	1	0.33
Traffic Accident	37	15	14	
(n and %)	%73	%71	%74	
Other Causes	14	6	5	0.59
(n and %)	%27	%29	%26	
Duration	2,96±1,67(1-9)	3,38±2,92(1-14)	3,94±4,65(2-22)	
Age	31,21±10,76 (18-59)	30,94±10,89 (18-54)	30,56±11,34 (17-61)	0.99
MMSE	19,50±1,82 (16-24)	18,10±2,76 (14-22)	15,43±1,71 (13-18)	0.01*
Verbal Fluency	7,24±2,53(3-13)	5,77±2,12(3-10)	5,60±2,30 (3-9)	0.12
Clock Drawing	5,71±1,61(3-9)	5,00±0,70 (4-6)	4,25±1,25 (3-6)	0.18
FAB	7,35±2,17 (4-12)	5,67±2,30 (3-7)	5,80±2,28 (3-8)	0.26

Abbreviations: **Mild:** Mild stage dementia, **Mild-Moderate:** Mild-moderate stage dementia, **Moderate:** Moderate stage dementia, **Duration:** Duration between trauma and diagnosis of dementia, **FAB:** Frontal Assessment Battery

Table 2: Comparison of Laboratory According to Stages of Dementia

	Mild (n:51)	Mild-Moderate (n:21)	Moderate (n:19)	p
Vitamin B12	175,72±54,04 (80-352)	196,30±48,03 (109-294)	180,13±50,20 (129-263)	0.20
Thyroid Stimulating Hormone	1,45±1,38 (0,52-7,01)	1,51±0,48 (0,83-2,40)	1,40±0,57 (0,45-2,39)	0.97
Abnormal EEG (n-%)	11 %22	6 %29	5 %26	0.31
Diffuse Ground Arrhythmia (n and %)	3 %6	4 %19	2 %10	
Epileptic Activity (n and %)	8 %16	2 %10	3 %16	
Abnormal MRI (n and %)	15 %29	11 %52	10 %53	0.87
Frontal lesion on MRI (n and %)	9 %18	4 %19	3 %16	
Temporal lesion on MRI (n and %)	5 %9	5 %24	3 %16	
Frontotemporal lesion on MRI (n and %)	1 %2	2 %9	4 %21	

Abbreviations: **Mild:** Mild stage dementia, **Mild-Moderate:** Mild-moderate stage dementia, **Moderate:** Moderate stage dementia

DISCUSSION

PTD is commonly associated with cognitive dysfunction. There were discrepancies between the results about the correlation between MRI and neurocognitive tests in previous studies. Levin et al. found a correlation between lesion localization and memory and planning⁽¹⁵⁾. In the present study, no difference was found between the stages of dementia and neurocognitive tests, except for MMSE and laboratory findings (MRI, EEG, Vitamin B12, TSH).

MMSE is a scanning test for dementia and it could be used for evaluating the orientation, complex attention, memory,

language and visuo-spatial ability⁽¹¹⁾. In this study, a significant difference ($p<0.01$) was found between the MMSE and stages of PTD. Furthermore, a negative correlation was observed between MMSE and GDS ($p<0.001$, $r: -0.474$). Patients, whose stages were classified as mild, lost points from the part of orientation and complex attention of MMSE, while those in the mild-moderate and the moderate groups also lost points from the part of memory. Memory loss can be observed due to damage to the temporal lobe. In the present study, despite the finding of no significant difference between the stages and MRI, the rate of mild-stage dementia patients with temporal lesions was lower than the other dementia groups (Table 2).

These findings suggest that the possible use of combined MRI and MMSE could play a role in the staging of PTD.

In the literature, the common etiology of head trauma is falling down from a height in children and elderly patients, while in adults, this is traffic accidents. In our study, the most frequent etiology of head trauma was traffic accidents, consistent with the literature, since our population comprised adult patients. Furthermore, most of the patients were male, consistent with the literature⁽⁸⁾.

Memory loss, personality changes, disinhibition, aggression, agitation, depression, abulia and apathy may be observed in PTD patients depending on the affected area⁽³⁾. Datta et al. demonstrated that headache and loss of concentration were the most common symptoms of PTD⁽⁴⁾. In this study, the most common presenting symptoms of PTD were amnesia, loss of motivation and concentration (n:68), consistent with the literature.

In general, focal findings are not observed on the neurological examinations of PTD cases⁽³⁾. In this study, only six patients, who were in moderate stage, had hemiparesis due to a parenchymal lesion. Hemiparesis can be a prognostic factor for PTD, but it this requires more comprehensive studies about PTD.

Hofman et al. demonstrated abnormal MRI findings in 12 of 21 (57%) patients after head trauma⁽⁷⁾, whereas Datta et al. showed abnormal MRI findings in 11 of 20 patients⁽⁴⁾ (55%). In both studies, most of the lesions were located in the frontal area, followed by the temporal areas. In the present study, 36 of 91 patients (40%) had abnormal MRI findings and the brain lesions were primarily in the frontal (n:16) areas and secondly, in the temporal areas (n:13), consistent with the literature (Table 2).

In the literature, non-epileptic abnormalities were observed at a rate of

75% in the acute period on EEG, and in the following period, these abnormalities normalized by 90% in the first year⁽¹²⁾. The duration for approval from the patients was more than one year and non-epileptic abnormalities were seen in only 10 % of the patients, consistent with the literature (Table 2).

In the literature, the normal range of vitamin B12 levels is still controversial; the mean level of vitamin B12 was determined as >300 pg/mL in healthy persons in one study⁽²²⁾, but in our study, the patients' mean level of vitamin B12 was <200 pg/mL (Table 1). Low vitamin B12 could facilitate the development of dementia. It is considered that this condition may have resulted from loss of daily activities, and that the patients could not provide sufficient self-care and nutritional intake.

The limitations of this study include the lack of the Glasgow Coma Scale's (GCS) of the patients at the emergency room on the first day of admission. If we knew the score of GCS on admission, we could comment on the relationship between GCS and the stages of PTD.

CONCLUSION

In the present study, a significant difference was found between the MMSE and the stages of PTD. We also found that the rate of mild stage dementia in patients with temporal lesions was lower than that of the other dementia groups. These findings suggest that the possible use of combined MRI and MMSE could play a role in the staging of PTD.

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